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Microstructure and Properties of $Al_{0.5}CoCrCuFeNiTi_x$ ($x = 0$–2.0) High-Entropy Alloys

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High-entropy $Al_{0.5}CoCrCuFeNiTi_x$ alloys are synthesized using the well-developed arc-melting and casting method. The molar ratio (x) of titanium is varied from 0 to 2.0. The microstructure, hardness and wear resistance of the alloys are investigated. The alloys exhibit simple FCC, BCC, CoCr-like and Ti_2Ni-like phases. For a small addition of titanium, the alloys form a monolithic FCC solid-solution phase. Two phases of β_1 and β_2 based on BCC appear at the titanium content of $x = 0.4$ and the β_1 phase becomes ordered at $x = 1.4$. With the increase of titanium content, copper segregates to the interdendrite region in which nano-precipitates form. A CoCr-like phase forms when x ranges from 0.8 to 1.2. Ti_2Ni-like phase forms when the titanium content exceeds $x = 1.0$. The hardness value increases with titanium content. The alloys with lower titanium content exhibit similar wear resistance to $Al_{0.5}CoCrCuFeNi$. The wear resistance is rapidly improved at titanium contents from 0.6 to 1.0, and reaches a maximum at $x = 1.0$. This is followed by a gradual decrease with further increase of the titanium. The mechanisms behind the strengthening and wear resistance of the alloys are discussed. [doi:10.2320/matertrans.47.1395]

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1. Introduction

Many past and present alloy systems are based on one principal matrix element with the incorporation of other elements for property and/or processing enhancement.[1,2] Metallurgists have generally confined themselves to systems where the number of alloying elements was limited in any one phase. This was partly because alloys made with an increasing number of component metals, many intermetallic compounds and complex microstructures form and the database necessary in terms of phase diagram information increases sharply.[3,4] However, this **impasse** has recently been broken. Metals have been mixed in a multi-metallic cocktail to make bulk metallic glasses, multi-functional, superelastic and superplastic alloys and nano-structured high entropy alloys.[3] High-entropy alloys were developed in recent years by Yeh *et al.*[3,5] Nano-structured high-entropy alloys with multiple principal elements are of equimolar or near-equimolar ratios and concentrations between 35 and 5 at%. Solid solutions with multi-principal elements will tend to be more stable at elevated temperatures because of their large mixing entropies.[6] Consequently, alloys containing a higher number of principal elements will more easily yield the formation of random solid solutions during solidification, rather than intermetallic compounds or other complicated phases.[5,7–9] $AlCoCrCu_{0.5}Ni$, $Al_xCoCrCuFeNi$, AlCoCrCuFeMoNiTiVZr, $AlCoCrFeMo_{0.5}NiSiTi$ and $AlCrFeMo_{0.5}NiSiTi$ high-entropy alloys are examples exhibiting a quite simple as-cast microstructure and promising performance.[7–11]

Among the previous alloys investigated, $Al_{0.5}CoCrCuFeNi$ alloy is an eminent alloy for structural applications, which possesses simple FCC structure and good hot and cold workability. Moreover, it exhibits a wear resistance close to SKD 61 steel with a hardness of 567 HV due to its large work hardening capacity although its as-cast hardness is 223 HV.[11] However, in accordance with the broadness of high-entropy-alloy design concept, the potential of $Al_{0.5}CoCrCuFeNi$ alloy can be explored further with suitable composition modification. In consideration of the strong bonding between titanium and many metal elements, titanium addition to the $Al_{0.5}CoCrCuFeNi$ alloy shall be investigated in the present study in order to improve its mechanical performance further.

2. Experimental Details

The $Al_{0.5}CoCrCuFeNiTi_x$ high-entropy alloys, with titanium molar ratios (x) of 0 to 2.0, were prepared by the arc melting and casting method. Refer to Ref. 7) for further details of this preparation method. The polished, *aqua regia* etched specimens were observed with an optical microscope and scanning electron microscope (SEM, JEOL JSM-5410). The chemical compositions were analyzed by SEM energy dispersive spectrometry (EDS). The EDS spot size was $1 \times 1\,\mu m^2$ and each specimen was examined at 5 points. An X-ray diffractometer (XRD, Rigaku ME510-FM2, Tokyo, Japan) was used for identification of the crystalline structure, with the 2θ scan ranging from 20 to 100 degrees at a rate of 1 degree/min. The radiation was from a 30 kV, 20 mA copper target. Hardness measurements were conducted using a Vickers hardness tester (Matsuzawa Seiki MV-1) under a load of 49 N and at a loading speed of 70 µm/s for 20 s. Based on five hardness measurements, the scattering error was found to be within 3%. Wear resistance was measured with a modified pin-on-disk wear tester, without lubrication, at a load of 29.4 N. Further details of the modified pin-on-disk wear tester are given in Ref. 8) The sliding velocity of wear-

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test sand belts, with 149-μm alumina (Al_2O_3) particles, was 0.5 m/s for a sliding distance of 20 m. Test pins, with a diameter of 8 mm, were controlled to move perpendicular to the sand-belt motion in order to avoid repetitive wear tracks and obtain reproducible data. The weight loss of the pins after wear tests was measured. It was divided by the alloy density and the sliding distance to evaluate the wear resistance.

3. Results and Discussion

3.1 XRD patterns for $Al_{0.5}CoCrCuFeNiTi_x$ alloys

The X-ray diffraction patterns of the $Al_{0.5}CoCrCuFeNiTi_x$ alloys are shown in Fig. 1. The crystalline structures of the as-cast $Al_{0.5}CoCrCuFeNiTi_x$ alloys are found to consist of FCC, BCC, CoCr-like (s), and Ti_2Ni-like (TL) phases. The $Al_{0.5}CoCrCuFeNiTi_{0.2}$ alloy possesses a FCC (a) structure, similar to that of $Al_{0.5}CoCrCuFeNi$, with a lattice constant of 0.362 nm. Two BCC phases (β_1 and β_2) appear at $x = 0.4$, and there is ordering in the β_1 phase at $x = 1.4$. The lattice constants of β_1 and β_2 are 0.287 and 0.292 nm, respectively. CoCr-like and Ti_2Ni-like phases form when x is in the ranges of 0.8 to 1.2 and 1.2 to 2.0, respectively. Both CoCr-like and Ti_2Ni-like phases are multi-element solid solutions. The CoCr phase, referred to as the s phase, has a tetragonal structure with lattice constants of $a = 0.885$ and $c = 0.459$ nm. Similar structures, with slightly different lattice constants, are found for s-FeCr and s-FeMo phases according to the JCPDS card. Ti_2Ni-like phase has a cubic structure with a lattice constant of 1.128 nm. From related binary and ternary phase diagrams,[12,13] it is found that s phase has a large solubility for Fe and Ni, and about 10 mass% solubility for Ti. On the other hand, the Ti_2Ni phase has solubility of up to 33, 20 and 30 mass% for Co, Cr and Fe, respectively. This means that both phases are composed of multi-principal elements.

3.2 Microstructures of $Al_{0.5}CoCrCuFeNiTi_x$ alloys

Figure 2 shows the SEM microstructures of the as-cast $Al_{0.5}CoCrCuFeNiTi_x$ alloys with different titanium contents. Typical cast dendrite and interdendrite structures (defined as DR and ID in the Figures, respectively) are observed in these alloys. The dendrites are divided into two zones, defined as DR-A and DR-B in Fig. 2. With low titanium contents, *i.e.* $x = 0.2$, both the dendrites and interdendrites are FCC with similar lattice constants to each other, as inferred from Fig. 1. Numerous nano-particles, with a size of 50 nm, can be seen in DR-B zone. When the titanium content (x) ranges from 0.4 to 0.6, DR-A is composed of FCC and β_1 phases, while DR-B is composed of FCC and β_2 phases. As the titanium content ranges from 0.8 to 1.0, DR-A consists of FCC, β_1 and CoCr-like phases while zone DR-B consists of FCC and β_2 phases. DR-B decomposes spinodally (defined as SD) when x ranges from 0.6 to 1.0. Similar spinodal decomposition with different morphologies in a BCC phase has been reported and discussed for $Al_yCoCrCuFeNi$ alloys ($y = 0$ to $y = 3.0$).[10] At a titanium content of $x = 1.2$, DR-A consists of FCC, β_1 and CoCr-like phases while DR-B consists of FCC, β_2 and Ti_2Ni-like phases. When the titanium content exceeds $x = 1.2$, DR-A is composed of FCC and β_1 phases while DR-B is composed of FCC, β_2 and Ti_2Ni-like phases.

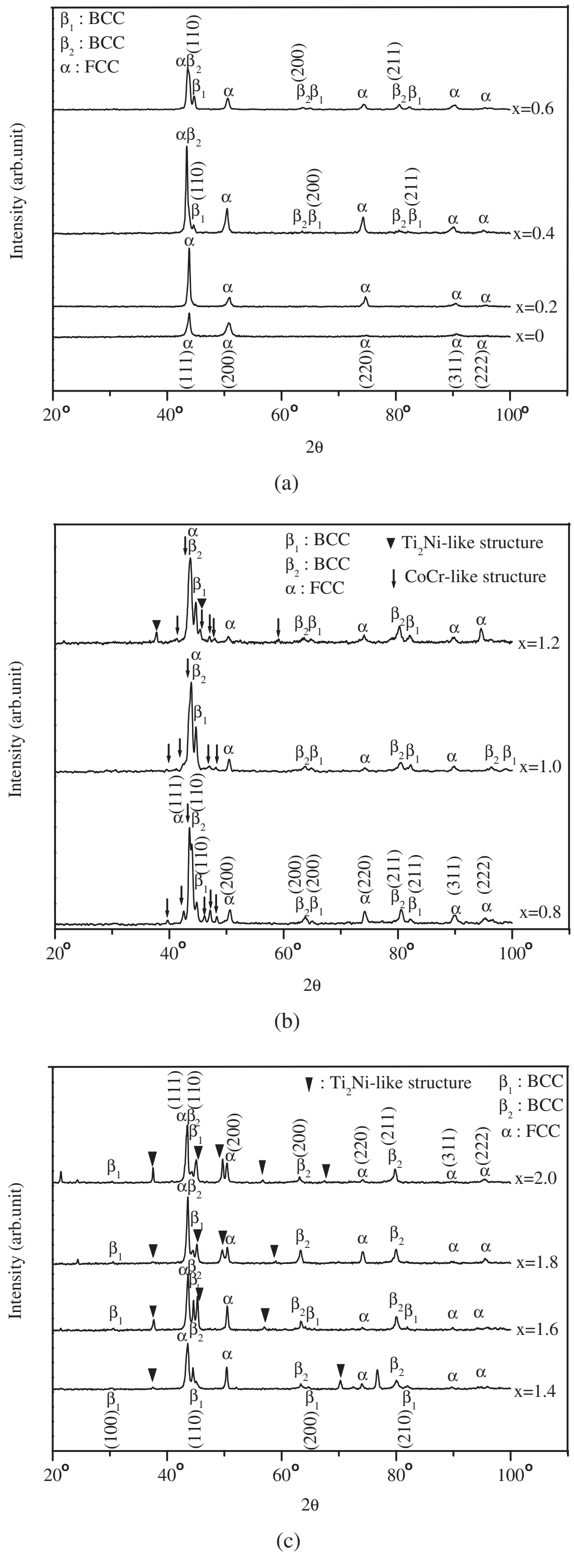

Fig. 1 XRD analyses of $Al_{0.5}CoCrCuFeNiTi_x$ alloys with different titanium contents (x value in molar ratio), (a) $x = 0$ to 0.6; (b) $x = 0.8$ to 1.2; (c) $x = 1.4$ to 2.0.

With increasing the titanium content from $x = 0.2$ to 1.0, the volume fraction of the interdendrite phase decreases. Beyond titanium contents of 1.0, the amount of interdendrite phase is found to increase again. The interdendrite is composed mainly of FCC phase because it contains copper ranging from 61 at% to 81 at%, nickel ranging from 13 at% to

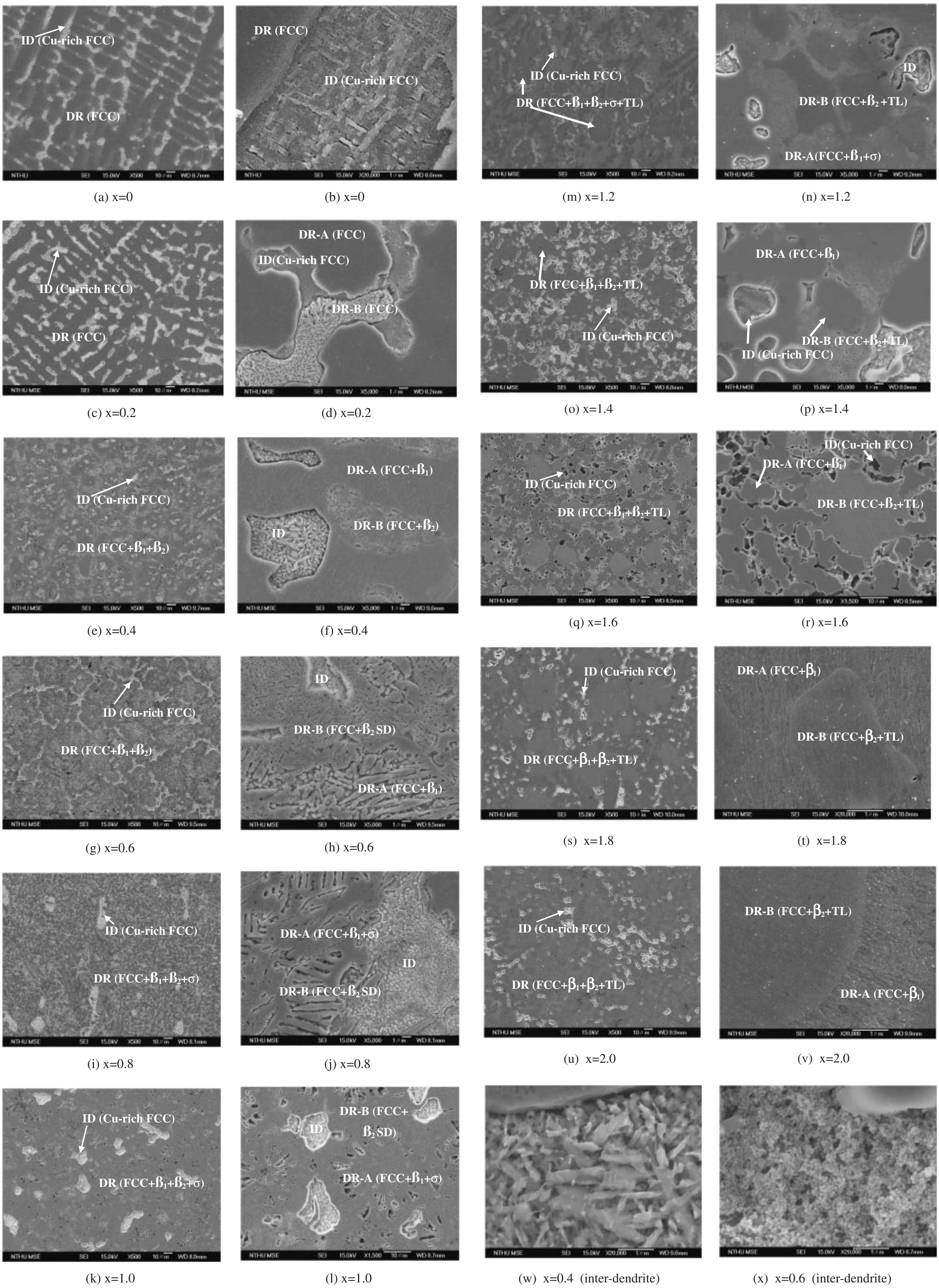

Fig. 2 SEM microstructures of as-cast $Al_{0.5}CoCrCuFeNiTi_x$ alloys with different titanium contents (x value), (a) and (b) 0, (c) and (d) 0.2, (e) and (f) 0.4, (g) and (h) 0.6, (i) and (j) 0.8, (k) and (l) 1.0, (m) and (n) 1.2, (o) and (p) 1.4, (q) and (r) 1.6, (s) and (t) 1.8, (u) and (v) 2.0, (w) 0.4 (inter-dendrite), (x) 0.6 (inter-dendrite), (DR: dendrite, ID: inter-dendrite, SD: spinodal decomposition, TL: Ti_2Ni-like phase, s: CoCr-like phase).

Table 1 Chemical compositions of cast $Al_{0.5}CoCrCuFeNiTi_x$ alloys in atomic percentage.

Alloy		Al	Cr	Cu	Fe	Co	Ni	Ti
$Al_{0.5}CoCrCuFeNi$	Nominal	9.09	18.18	18.18	18.18	18.18	18.18	—
	Dendrite	9.4	19.5	12.6	19.4	18.7	20.4	—
	Inter-dendrite	12.5	4.1	61.3	4.7	5.2	12.2	—
$Al_{0.5}CoCrCuFeNiTi_{0.2}$	Nominal	8.7	17.5	17.5	17.5	17.5	17.5	3.5
	Dendrite-A	7.3	21.6	10.5	20.8	20.0	17.1	2.7
	Dendrite-B	19.1	9.6	14.4	10.0	15.0	22.8	9.1
	Inter-dendrite	11.4	3.3	64.6	4.3	4.4	10.5	1.5
$Al_{0.5}CoCrCuFeNiTi_{0.4}$	Nominal	8.4	16.9	16.9	16.9	16.9	16.9	6.7
	Dendrite-A	5.5	23.3	7.6	22.9	19.9	14.6	6.2
	Dendrite-B	12.1	16.0	10.2	14.6	17.3	20.6	9.2
	Inter-dendrite	11.1	3.6	60.7	5.1	5.3	13.2	1.0
$Al_{0.5}CoCrCuFeNiTi_{0.6}$	Nominal	8.1	16.3	16.3	16.3	16.3	16.3	9.8
	Dendrite-A	5.9	25.6	8.0	21.4	17.2	13.4	8.5
	Dendrite-B	9.9	14.6	11.7	15.6	16.9	18.5	12.8
	Inter-dendrite	6.7	2.8	68.9	4.8	4.0	10.2	2.6
$Al_{0.5}CoCrCuFeNiTi_{0.8}$	Nominal	7.9	15.8	15.8	15.8	15.8	15.8	12.6
	Dendrite-A	7.4	12.8	11.2	15.9	17.0	18.6	17.1
	Dendrite-B	18.8	5.2	6.9	7.3	20.2	20.6	21.0
	Inter-dendrite	5.1	2.0	77.5	3.3	2.6	6.9	2.6
$Al_{0.5}CoCrCuFeNiTi_{1.0}$	Nominal	7.6	15.3	15.3	15.3	15.3	15.3	15.3
	Dendrite-A	13.8	17.8	5.9	14.1	18.2	14.8	15.4
	Dendrite-B	7.0	12.0	11.0	16.5	17.2	18.4	17.9
	Inter-dendrite	5.5	2.1	77.4	3.4	2.5	6.7	2.4
$Al_{0.5}CoCrCuFeNiTi_{1.2}$	Nominal	7.4	14.9	14.9	14.9	14.9	14.9	17.9
	Dendrite-A	1.7	15.4	2.1	24.3	19.0	9.8	27.7
	Dendrite-B	18.3	4.6	6.3	7.1	20.4	18.5	24.8
	Inter-dendrite	3.9	2.8	73.7	4.2	3.1	8.2	4.1
$Al_{0.5}CoCrCuFeNiTi_{1.4}$	Nominal	7.2	14.4	14.4	14.4	14.4	14.4	20.2
	Dendrite-A	1.9	18.4	2.5	23.1	17.2	8.8	28.1
	Dendrite-B	18.5	4.3	6.4	7.6	19.5	18.2	25.5
	Inter-dendrite	5.6	2.6	74.2	3.0	2.2	7.9	4.5
$Al_{0.5}CoCrCuFeNiTi_{1.6}$	Nominal	7.0	14.1	14.1	14.1	14.1	14.1	22.5
	Dendrite-A	1.5	18.5	3.3	22.4	15.7	10.5	28.1
	Dendrite-B	17.5	4.6	6.8	7.7	18.8	17.5	27.1
	Inter-dendrite	3.8	2.4	77.7	3.1	2.1	7.6	3.3
$Al_{0.5}CoCrCuFeNiTi_{1.8}$	Nominal	6.8	13.7	13.7	13.7	13.7	13.7	24.6
	Dendrite-A	1.9	21.7	2.4	21.3	15.0	8.4	29.3
	Dendrite-B	15.5	4.6	6.8	9.0	18.6	16.2	29.3
	Inter-dendrite	2.7	1.8	81.2	2.3	1.5	6.5	4.0
$Al_{0.5}CoCrCuFeNiTi_{2.0}$	Nominal	6.6	13.3	13.3	13.3	13.3	13.3	26.6
	Dendrite-A	2.1	23.0	2.6	20.9	13.7	7.4	30.3
	Dendrite-B	16.1	4.2	8.5	7.1	16.3	18.7	29.1
	Inter-dendrite	3.6	2.3	80.9	1.9	1.9	4.6	4.8

4.6 at%, and aluminum ranging from 12.5 at% to 2.7 at% for titanium contents of 0 to 2.0. According to the ternary phase diagram of the Al-Cu-Ni system, the primary phase of the investigated region is Cu-rich FCC phase with the Ni_3Al-type phase for low aluminum content, or NiAl-type phase for higher aluminum contents, in addition to the other minor phases.[13] The precipitates in the interdendrite regions show different morphologies with increasing titanium contents. The precipitates are needle-like in the titanium content range of 0.2 to 0.4, whereas they appear as nano-particles at higher titanium contents. The size of precipitates increases from 20 nm to 150 nm with increasing titanium content. Figure 2 (w) and (x) is the typical morphology of the precipitate. A high concentration of other elements may account for the nano-precipitation due to their sluggish cooperative diffusion.[7,10)]

The chemical compositions of the cast alloys, analyzed by EDS, are summarized in Table 1. Although copper segregation is observed for the interdendrite regions, the majority dendrite is basically composed of multi-principal elements at concentrations comparable to the nominal, except for copper. With the increase of titanium content, more and more copper

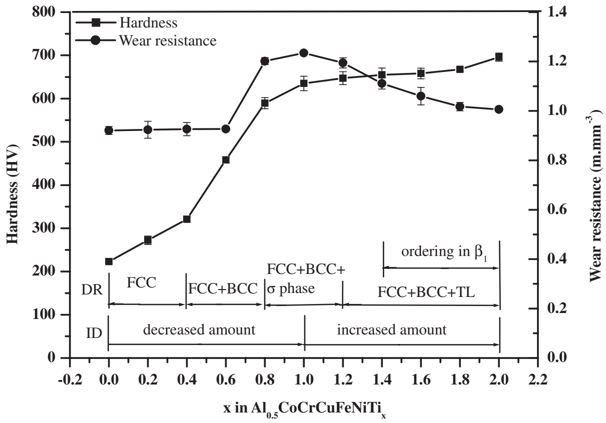

Fig. 3 Vickers hardness and wear resistance of $Al_{0.5}CoCrCuFeNiTi_x$ alloys with different titanium contents.

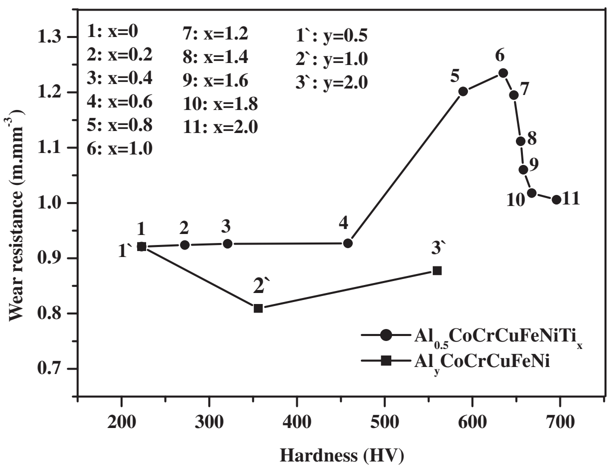

Fig. 4 Wear resistance of $Al_{0.5}CoCrCuFeNiTi_x$ and $Al_yCoCrCuFeNi$ alloys as a function of hardness.

segregates to the interdendrite region. This is attributed to the mixing enthalpies between copper and aluminum, cobalt, chromium, iron, nickel, and titanium which are −1, 6, 12, 13, 4, and −9 kJ/mole, respectively, whereas the mixing enthalpies between titanium and aluminum, cobalt, chromium, iron, and nickel are −30, −28, −7, −17, and −35 kJ/mole, respectively.[14] Thus, aluminum, cobalt, chromium, iron, and nickel attract titanium to the dendrite region and copper acts to exclude cobalt, chromium, iron, and nickel. It is also noted that copper concentrations in all interdendrite regions are higher than 60 at% and the concentration of aluminum and nickel are approximately 3 at% to 13 at%. This is considered to be a result of the high chemical affinity that copper has with aluminum and nickel compared with cobalt, chromium, iron and titanium. The chemical analysis of DR-A and DR-B refer to the identification of the crystal structure. The concentrations of the larger Al and Ti atoms in DR-B are higher than that of DR-A. However the reverse is true for the smaller Co, Cr and Fe atoms, for which DR-A has the greater concentration of these elements. In accordance with XRD analysis (Fig. 1), where the lattice constant of β_1 is smaller than that of β_2, it is concluded that the BCC phases present in DR-A and DR-B are β_1 and β_2, respectively. Since the contents of nickel and titanium are higher in DR-B than DR-A, it is considered that Ti_2Ni-like phase is formed in DR-B. The chromium concentration in DR-A is found to be greater than that in DR-B. Moreover, the concentration ratio between chromium and cobalt in DR-A is close to unity, unlike that in DR-B. Based on this it is suggested that a CoCr-like phase forms in DR-A.

3.3 Hardness of $Al_{0.5}CoCrCuFeNiTi_x$ alloys

Figure 3 shows the Vickers hardness of $Al_{0.5}$-$CoCrCuFeNiTi_x$ alloys as a function of titanium content. The hardness increases from 225 to 660 HV with increasing titanium content. The increase in hardness appears to occur in three stages, first a gradual increase in hardness occurs from 0 to 0.4, then a sharp increase from 0.4 to 1.0 followed by another gradual increase at the highest titanium contents. In comparison with the microstructural evolution shown in Fig. 2, it appears that the formation of BCC and CoCr-like phases is the most important factor of strengthening. The decreased volume fraction of interdendrite, as the titanium content is increased from 0 to 1.0, appears also to contribute. The solid solution hardening in the FCC and BCC phases becomes weaker when the titanium content exceeds 1.0. This is reasonable since in these solution phases there is no matrix element per se. Therefore, all of the atoms may be regarded as solute atoms, and as a result they are a highly concentrated solid solution and show insensitivity to the incorporation of other atoms with similar bonding strength.[5)]

The sharp increase in hardness beyond a titanium content of 0.4 is considered to be a result of the BCC phase that forms being stronger than the FCC phase.[5,11)] This can be explained with the following viewpoints. The first is that slip along the closest packing planes {110} in BCC is more difficult than the {111} closest packing planes in FCC. This is because {110} planes in BCC are less dense resulting in a smaller interplanar spacing and higher lattice friction for dislocation motion.[15–17)] The second is due to the incorporation of stronger binding elements like Al and high melting point elements like Cr and Ti in the BCC phase that increases the Young's modulus and the slip resistance.[16,17)] Another is the spinodal decomposition of BCC phase, wherein a nano-spaced spinodal structure might provide a nano-composite strengthening effect.[18)]

3.4 Wear resistance of $Al_{0.5}CoCrCuFeNiTi_x$ alloys

The effect of titanium content on the wear resistance of the $Al_{0.5}CoCrCuFeNiTi_x$ alloy is shown in Fig. 3. The wear resistance of $Al_{0.5}CoCrCuFeNiTi_x$ is similar to that of $Al_{0.5}CoCrCuFeNi$ for titanium contents in the range of 0 to 0.6. As the titanium content increases from 0.6 to 0.8, the wear resistance rapidly improves from 0.93 m/mm^3 to 1.20 m/mm^3. The wear resistance reaches the maximum of 1.24 m/mm^3 at $x = 1.0$ and then decreases at greater titanium contents. Figure 4 shows the relationship between wear resistance and hardness. From the wear resistance analysis shown in Figs. 3 and 4, it is found that the wear resistance of $Al_{0.5}CoCrCuFeNiTi_x$ alloys is seemingly independent to their hardness. In general, the wear resistance of materials is proportional to the Vickers' hardness.[19)] However, the wear resistance of $Al_{0.5}CoCrCuFeNiTi_x$ alloys does not follow the

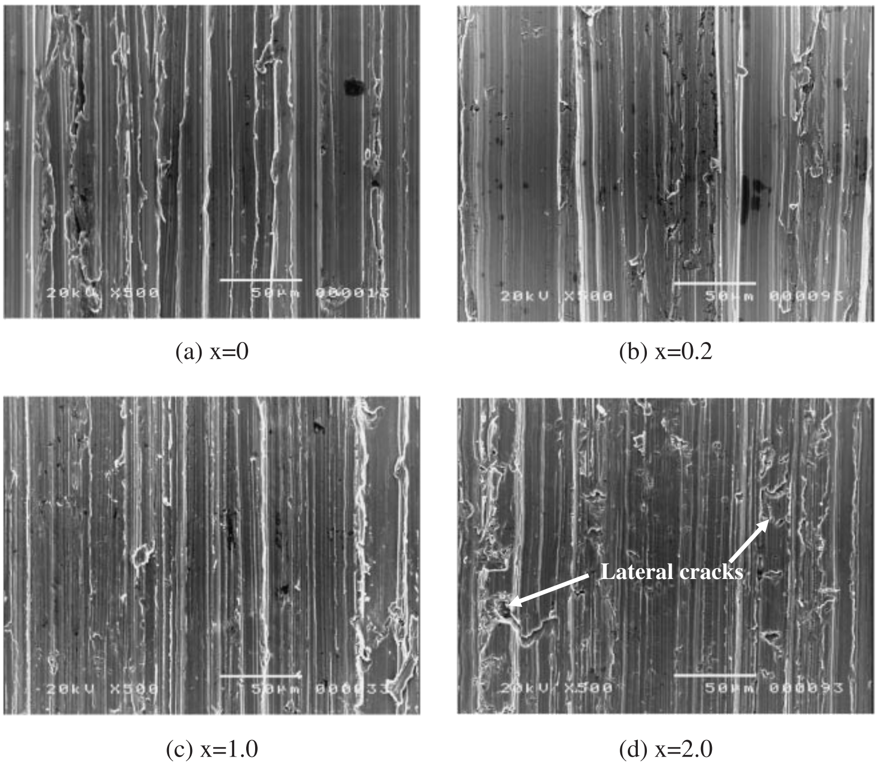

Fig. 5 Surface morphologies of $Al_{0.5}CoCrCuFeNiTi_x$ alloys with different titanium contents (x value) after wear tests, (a) 0, (b) 0.2, (c) 1.0, (d) 2.0.

rule as seen in Fig. 4. The wear resistance changes very little as the hardness value increases from 225 HV to 460 HV. It has a great increase at the hardness values of 460 HV to 660 HV, and then rapidly drops when the hardness value exceeds 660 HV.

The wear resistance of $Al_yCoCrCuFeNi$ alloys (y value in molar ratio) for different aluminum content from previous work by the authors[11] is also shown in Fig. 4 for comparison. The study reported that $Al_{0.5}CoCrCuFeNi$ alloy is simple FCC and exhibits high work-hardening that results in a significant surface-hardening phenomenon and good performance during abrasion wear. Its wear resistance is even close to that of SKD61 steel with a hardness of 567 HV. However, more Al addition ($y = 1$ and 2) yields more BCC phase, which increases the hardness but degrades the wear resistance. The reason is attributed to the more brittle behavior due to the formation of an ordered BCC phase as determined from the microstructure analysis, mechanical testing and worn surface.[10,11] On the other hand, the addition of titanium is more effective than aluminum addition in increasing both hardness and wear resistance. As the content of the addition elements exceeds 0.5, the contribution of increasing titanium content to wear resistance is much better than that of increasing aluminum content. This is related to two microstructural factors: (1) the different ordering effect on BCC phase exerted by aluminum and titanium; and (2) the formation of a CoCr-like phase. With regard to the former, aluminum tends to render BCC phase more highly ordered as revealed by the ordered BCC peak at 31° in the XRD patterns of $Al_yCoCrCuFeNi$, when y exceeds 1.0.[10,11] However, ordering of BCC phase in $Al_{0.5}CoCrCuFeNiTi_x$ alloys only begins at $x = 1.4$ as mentioned in Section 3.1. This might explain why the low titanium content gives better resistance than that of $Al_yCoCrCuFeNi$ alloys with $y = 1$ and 2. The second factor, *i.e.* the formation of a much stronger CoCr-like phase between $x = 0.8$ and 1.2, not only gives a higher hardness but also significantly improves the wear resistance in a similar titanium content range. This benefit becomes diminished when CoCr-like phase is gradually replaced by Ti_2Ni-like and ordered BCC phases with more titanium addition. Therefore, compared with $Al_yCoCrCuFeNi$, a proper titanium addition to the $Al_{0.5}CoCrCuFeNi$ alloys promotes the alloy performance further, revealing a higher potential for the application to tool alloy industry.

Figure 5 shows the worn surface of the alloys with different titanium contents. When the titanium content is less than 1.0, no lateral cracks are seen at the surface, thus revealing the high ductility of these alloys. This implies that all the phases are tough enough to resist the impacts with coarse abrasives, of around 150 µm in size. Moreover, ductile deformation along the plough lines is seen to be more significant in the alloys with lower titanium content and hardness level. In contrast, many lateral cracks are found on the worn surfaces (Fig. 5(c)) after the titanium content exceeds 1.0 indicating the deformation behavior tends to be of a more brittle nature. This concurs with the decrease in wear resistance beyond $x = 1.0$. Moreover, this reflects the effectiveness of strong CoCr-like phase in improving wear resistance and hardness as compared with Ti_2Ni-like and ordered BCC phase, since the former phase is gradually replaced by the latter two as the titanium content is increased. It can be seen that the wear resistance at $x = 2.0$ drops towards the level of the $Al_yCoCrCuFeNi$ alloys ($y = 1$ and 2), which also have an ordered BCC phase.

4. Conclusions

The $Al_{0.5}CoCrCuFeNiTi_x$ alloys exhibited quite simple FCC, BCC, CoCr-like and Ti_2Ni-like phases. With small titanium addition, the alloys had a monolithic FCC solid-solution phase. Two BCC phases of β_1 and β_2 appeared at a titanium content of $x = 0.4$ and the β_1 phase became ordered at $x = 1.4$. As the titanium content was increased, copper segregated to the interdendrite region, in which nano-precipitates form. CoCr-like and Ti_2Ni-like phases form when x is in the ranges of 0.8 to 1.2 and 1.2 to 2.0, respectively. The hardness value increased with titanium content as stronger BCC and CoCr-like phases formed in the dendrite regions. The alloys with lower titanium content exhibited similar wear resistance to $Al_{0.5}CoCrCuFeNi$. The wear resistance was rapidly improved with increasing the titanium content from 0.6 to 1.0 and reached a maximum at $x = 1.0$ followed by gradual decrease at higher titanium contents. The better combination of the wear resistance and hardness, as compared with those of the alloys with high aluminum addition, was attributed to the formation of the CoCr-like phase between $x = 0.8$ and 1.2 and the suppression of the ordered BCC phase.

Acknowledgements

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